

# A Magnification-Dependent Visual Quality Index for Spherical Newton Mirrors

Derivation, Implementation, and Validation of  $Q_{\text{vis}}(V)$  in the Newton Mirror Calculator v5.0

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A Newton telescope with a spherical primary mirror generates spherical aberration, the visual impact of which depends heavily on the magnification used. Existing quality metrics, such as the Strehl ratio or MTF shape loss, are independent of magnification and therefore fail to describe what the observer actually perceives. We derive a visual quality index  $Q_{\text{vis}}(V)$  that calculates the CSF-weighted MTF integral of the spherical mirror relative to a diffraction-limited parabolic mirror.  $Q_{\text{vis}}(V) = 1$  denotes no perceptible degradation;  $Q_{\text{vis}}(V) \rightarrow S_{\text{exakt}}$  at very high magnifications. We discuss an analytical approximation formula  $\tilde{Q}(V)$ , its limitations, and a critical implementation error (a factor-of-2 error during  $W_b$  inversion) that has been resolved in v5.0.1.

## 1 Introduction and Motivation

The quality comparison between spherical and parabolic Newton mirrors is traditionally conducted using the Strehl ratio  $S$ . A mirror is considered "diffraction-limited" for  $S \geq 0.80$  (Rayleigh criterion) or optically excellent for  $S \geq 0.95$ . However, both boundaries are independent of magnification—they provide no information regarding the specific magnification at which a quality difference actually becomes visible.

Suiter [1] introduced the critical magnification  $V_{\text{krit}} = D \cdot 0.7 \cdot \sqrt{S}$ : below this threshold, the resolution limit of the eye dominates, rendering the spherical aberration imperceptible. Above  $V_{\text{krit}}$ , the aberration blur becomes visible. While this criterion is qualitatively correct, it does not provide a continuous quality curve.

The approach described here replaces the binary threshold with a continuous index  $Q_{\text{vis}}(V)$  based on the CSF-weighted MTF integral. It is closely related to the "Visible Contrast Loss" concept in applied optics [2, 3] and has been adapted for practical astronomical observation.

## 2 Wavefront and Strehl Ratio

### 2.1 Spherical Aberration

A spherical mirror with aperture  $D$ , focal length  $f$ , and observation wavelength  $\lambda$  generates a PtV (Peak-to-Valley) wavefront error at the paraxial focus:

$$W_p = \frac{D^4}{1024 \cdot f^3 \cdot \lambda} \quad [\text{in units of } \lambda] \quad (1)$$

Optimal defocusing (balancing the  $\rho^4$  term with a  $\rho^2$  defocus) minimizes the RMS wavefront error. The resulting wavefront at best focus is given by:

$$W(\rho) = 8 \cdot W_b \cdot (\rho^4 - \rho^2) \quad \text{with } W_b = \frac{W_p}{4} \quad (2)$$

The RMS wavefront error for this profile is  $W_{\text{rms}} = \frac{W_b}{1.5 \cdot \sqrt{5}}$ .

### 2.2 Strehl Ratio: Approximation vs. Exact Value

The Maréchal approximation yields the Strehl ratio directly from  $W_{\text{rms}}$ :

$$S_{\text{Mar}} \approx \exp(-(2\pi \cdot W_{\text{rms}})^2) \quad [\text{valid for } W_{\text{rms}} < 0.07\lambda] \quad (3)$$

For fast mirrors ( $f/D < 8$ , corresponding to  $W_b > 0.08\lambda$ ), the Maréchal approximation significantly overestimates the true Strehl ratio. The exact value must be derived from the pupil integral:

$$\begin{aligned} S_{\text{exakt}} &= |\langle \exp(i \cdot 2\pi \cdot W(\rho)) \rangle|^2 \\ &= \left( 2 \int_0^1 \cos(2\pi \cdot W_c(\rho)) \rho \, d\rho \right)^2 + \left( 2 \int_0^1 \sin(2\pi \cdot W_c(\rho)) \rho \, d\rho \right)^2 \end{aligned} \quad (4)$$

where  $W_c(\rho) = W(\rho) - \langle W \rangle$  (piston-removed).

In the software program, equation (4) is evaluated numerically using  $n = 2000$  sampling points (`_strehl_exakt(Wb)`). From v5.0.0 onwards,  $S_{\text{exakt}}$  is used consistently for all perception-relevant calculations.

## 3 Modulation Transfer Function

### 3.1 MTF of the Parabolic Mirror

The MTF of a diffraction-limited circular telescope (parabolic mirror) possesses the analytical form:

$$m_p(\nu) = \frac{2}{\pi} \cdot (\arccos(\nu) - \nu \cdot \sqrt{1 - \nu^2}) \quad \nu \in [0, 1] \quad (5)$$

The normalized spatial frequency  $\nu = 1$  corresponds to the diffraction cutoff frequency  $f_c = D/(\lambda \cdot 206265)$  [arcsec<sup>-1</sup>]. In image space, this corresponds to  $f_{c,\text{mm}} = D/(\lambda \cdot f)$  [Lp/mm].

### 3.2 MTF of the Spherical Mirror

The relative MTF of the spherical mirror  $m_{\text{sr}}(\nu, W_b)$  — normalized to  $m_{\text{sr}}(0) = 1$  — is calculated via the Talbot integral. It integrates the complex amplitude over all pairs of pupil points separated by a distance  $\nu$  in the pupil plane:

$$m_{\text{sr}}(\nu, W_b) = \frac{|\int \exp(i \cdot 2\pi \cdot \Delta W(x, \nu)) h(x, \nu) dx|}{\int h(x, \nu) dx} \quad (6)$$

$\Delta W(x, \nu) = W(x - \nu/2) - W(x + \nu/2)$  (wavefront difference)

$h(x, \nu)$  = geometric overlap function of the circular pupil

In the program (`mtf_sph_rel`), the integral is evaluated numerically using  $n = 80$  sampling points and is LRU-cached. Two focus modes are implemented: the standard mode (Best Focus, Equation 2) and the paraxial focus  $W(\rho) = 8 \cdot W_b \cdot \rho^4$  for comparison with Sacek [4] and Born & Wolf [5].

## 4 Derivation of the Visual Quality Index $Q_{\text{vis}}(V)$

### 4.1 Motivation: Why Strehl and MTF Alone Are Insufficient

The Strehl ratio  $S$  and MTF  $m_{\text{sr}}(\nu)$  describe the physical properties of the telescope, but they are independent of both the magnification used and the human visual system. A spherical mirror with  $S = 0.85$  at  $V = 50\times$  is barely distinguishable from a parabolic mirror by the human eye; at  $V = 300\times$ , however, the quality difference becomes distinctly visible.

The underlying cause is the Contrast Sensitivity Function (CSF) of the eye: the human eye exhibits a bandpass-shaped sensitivity profile with a peak at approximately 10 cycles per degree (cpd). Spatial frequencies lying outside this peak are perceived with significantly lower fidelity.

### 4.2 Contrast Sensitivity Function (CSF)

We utilize the CSF defined by van Meeteren / Watson-Ahumada [2, 3]:

$$\text{CSF}(f) = 2.6 \cdot (0.0192 + 0.114 \cdot f) \cdot \exp(-(0.114 \cdot f)^{1.1}) \quad f \text{ in cpd} \quad (7)$$

This function peaks at around 10 cpd, drops sharply towards high frequencies (the resolution limit of the eye  $\sim 60$  cpd), and decreases gradually towards low frequencies (lateral inhibition). The conversion of the normalized telescope frequency  $\nu$  into cpd is achieved via the magnification  $V$ :

$$f_{\text{cpd}}(\nu, V) = \nu \cdot f_c \cdot \frac{3600}{V} \quad f_c \text{ in arcsec}^{-1}, V \text{ in } \times \quad (8)$$

At low magnifications ( $V \approx 30\text{--}50\times$ ), all telescope frequencies fall far to the right of the CSF peak where  $f > 50$  cpd and  $\text{CSF} \approx 0$ . The eye cannot resolve the aberrations, resulting in  $Q_{\text{vis}} \approx 1$  regardless of the mirror type. As  $V \approx V_{\text{krit}}$ , the telescope frequencies shift into the most sensitive region of the CSF, making aberrations maximally apparent.

### 4.3 Definition of $Q_{\text{vis}}(V)$

$Q_{\text{vis}}(V)$  is defined as the ratio of the CSF-weighted MTF integral of the spherical mirror to that of a diffraction-limited parabolic mirror at identical magnification:

$$Q_{\text{vis}}(V) = \frac{\int_0^1 m_{\text{sr}}(\nu) \cdot m_p(\nu) \cdot \text{CSF}(f_{\text{cpd}}(\nu, V)) d\nu}{\int_0^1 m_p(\nu) \cdot \text{CSF}(f_{\text{cpd}}(\nu, V)) d\nu} \quad (9)$$

This definition yields three core characteristics:

1. **Limit at low magnification:** For  $V \rightarrow V_{\min}$ ,  $\text{CSF}(f_{\text{cpd}})$  becomes extremely small for all  $\nu > 0$ , since all telescope frequencies lie far beyond the eye's resolving capabilities. The ratio is dominated by  $\nu \rightarrow 0$ , where  $m_{\text{sr}}(0) = 1 \rightarrow Q_{\text{vis}}(V_{\min}) \rightarrow 1$ . Thus, the spherical mirror is indistinguishable from a parabolic mirror at low magnifications.
2. **Limit at high magnification:** For  $V \rightarrow \infty$ , all telescope frequencies shift beneath the CSF peak ( $f_{\text{cpd}} \rightarrow 0$ ). Since  $\text{CSF}(f \rightarrow 0) \rightarrow \text{const.}$ , it approximately cancels out, and  $Q_{\text{vis}}$  approaches the CSF-weighted MTF shape loss—a value close to, but not identical with,  $S_{\text{exakt}}$ .
3. **No Strehl pre-factor:** Unlike older formulations, we do *not* multiply the absolute MTF  $m_{\text{sa}} = m_{\text{sr}} \cdot m_p \cdot S$  by  $S$ . The Strehl effect is already inherently factored into  $m_{\text{sr}}$ ; an additional factor of  $S$  would double-count the quality degradation.

## 4.4 Numerical Implementation

The integral (9) is computed using  $n = 200$  equidistant sampling points across  $\nu \in [0, 1]$  via the trapezoidal rule. The implementation inside `perceived_quality_exakt()` is structured as follows:

```
fc      = D / (lambda[mm] * 206265)      # Cutoff frequency [arcsec^-1]
nu      = linspace(0, 1, 200)
mp      = [mtf_para(nu_i)                for nu_i in nu]
msr     = [mtf_sph_rel(nu_i, Wb)         for nu_i in nu]
w_csf   = [csf(nu_i*fc*3600/V)           for nu_i in nu]

Q_vis   = trapz(msr * mp * w_csf, nu) / trapz(mp * w_csf, nu)
```

A variant named `perceived_quality_exakt_von_Wb(D, Wb,  $\lambda$ , V)` accepts  $W_b$  directly, bypassing back-calculation via  $D$  and  $f$ . This is utilized for the slider comparison curve, which simulates a hypothetical mirror with a target Strehl value  $S_{\text{slide}}$ .

## 5 Analytical Approximation Formula $\tilde{Q}(V)$

### 5.1 Derivation

For quick estimations and rapid computation, an analytical approximation is highly desirable. We formulate  $\tilde{Q}(V)$  as an interpolating function between two known boundary conditions:

For  $V \rightarrow V_{\min}$ :  $\tilde{Q} \rightarrow 1$  (no perceptible loss)

For  $V \rightarrow \infty$ :  $\tilde{Q} \rightarrow S_{\text{exakt}}$  (asymptote)

A straightforward damping function  $w(V)$  satisfying  $w(V_{\text{krit}}) = 0.5$  meets these requirements:

$$w(V) = \frac{1}{1 + (V_{\text{krit}}/V)^2} \quad \text{mit } V_{\text{krit}} = D \cdot 0.7 \cdot \sqrt{S_{\text{exakt}}} \quad (10)$$

$$\tilde{Q}(V) = S_{\text{exakt}} + (1 - S_{\text{exakt}}) \cdot (1 - w(V)) \quad (11)$$

The term  $(1 - w(V))$  interpolates between 0 (when  $V \ll V_{\text{krit}}$ ) and 1 (when  $V \gg V_{\text{krit}}$ ). At  $V = V_{\text{krit}}$ ,  $w = 0.5$ , which yields  $\tilde{Q}(V_{\text{krit}}) = (1 + S_{\text{exakt}})/2$ .

## 5.2 Limitations of the Approximation

While  $\tilde{Q}(V)$  qualitatively mirrors the magnification-dependent behavior correctly, it systematically overestimates perceived quality in the mid-magnification range by typically 1–3% because it does not model the bandpass shape of the CSF. Furthermore, unlike  $Q_{\text{vis}}(V)$ ,  $\tilde{Q}$  lacks a physically justified upper magnification limit.

Table 1: Comparison of Exact vs. Approximation for  $S = 0.90$ ,  $D = 355$  mm

| Magnification   | Exact $Q_{\text{vis}}$ | $\tilde{Q}$ | Difference |
|-----------------|------------------------|-------------|------------|
| $V = 50\times$  | 0.971                  | 0.981       | +0.010     |
| $V = 100\times$ | 0.950                  | 0.966       | +0.016     |
| $V = 200\times$ | 0.932                  | 0.958       | +0.026     |
| $V = 400\times$ | 0.887                  | 0.926       | +0.039     |

The program displays both curves superimposed within the perception diagram:  $Q_{\text{vis}}$  as a solid line (exact) and  $\tilde{Q}$  as a dashed line (approximation). The hatching between both curves visualizes the model residual.

## 6 Critical Implementation Bug: $W_b$ Inversion

### 6.1 The Factor-of-2 Error

To generate the slider comparison curve, the  $W_b$  value corresponding to a desired target Strehl  $S_{\text{slide}}$  must be isolated. Prior to v5.0.0, this inversion was performed using the Maréchal approximation:

$$W_{\text{rms}} = \frac{\sqrt{-\ln S}}{2\pi} \quad (12)$$

$$W_{b,\text{alt}} = W_{\text{rms}} \cdot 1.5 \cdot \sqrt{5} \quad (13)$$

Because the Maréchal approximation overestimates the true Strehl ratio for relevant Strehl values ( $S < 0.99$ ), its inverse function generates a value for a given  $S$  that is systematically too large. Numerical analysis revealed a scale factor error of roughly 2.001–2.016 across the entire operationally relevant domain:

Table 2: Factor-of-2 Error of the Maréchal Inversion

| $S_{\text{target}}$ | $W_{b,\text{alt}}$ (Maréchal) | $W_{b,\text{neu}}$ (Bisection) | Factor |
|---------------------|-------------------------------|--------------------------------|--------|
| 0.98                | 0.075875                      | 0.037910                       | 2.001× |
| 0.95                | 0.120900                      | 0.060340                       | 2.004× |
| 0.90                | 0.173275                      | 0.086315                       | 2.007× |
| 0.85                | 0.215203                      | 0.106988                       | 2.011× |
| 0.80                | 0.252167                      | 0.125105                       | 2.016× |

A  $W_b$  value that is twice as large as it should be introduces heavily exaggerated aberration into the MTF calculation. Consequently, the slider curve reflected the performance of a vastly inferior mirror. At  $S_{\text{slide}} = 0.90$  and  $V = 200\times$ , the error in  $Q_{\text{vis}}$  was approximately 0.18—a massive quality disparity that manifested in the diagram as a large divergence between the exact and approximate curves.

## 6.2 The Fix: Bisection Search via the Pupil Integral

Starting with v5.0.1,  $W_b$  is determined via a bisection search utilizing the exact pupil integral, ensuring that `_strehl_exakt(Wb) == S_target` is satisfied (computed over 60 bisection iterations, achieving a precision of  $< 10^{-18}$ ):

```
lo, hi = 0.0, 0.44
for _ in range(60):
    mid = (lo + hi) / 2
    if _strehl_exakt(mid) > S_target: lo = mid
    else: hi = mid
W_b = (lo + hi) / 2
```

Following this correction, the deviation between  $Q_{\text{vis}}$  and  $\tilde{Q}$  for the slider curve aligns precisely with the deviation observed for the standard spherical curve (1–4%), matching the expected structural residual of the approximation formula.

*Note:* The approximation formula  $\tilde{Q}$  must utilize the physically correct asymptote  $S_{\text{ex,ph}} = \text{\_strehl\_exakt}(W_{b,\text{neu}})$ , rather than the virtual slider input value  $S_{\text{slide}}$ . The legend text continues to display  $S_{\text{slide}}$  (the user’s desired Strehl), while the actual curve profile is driven by  $S_{\text{ex,ph}}$ .

## 7 Scope of Validity and Limitations

### 7.1 Magnification Range

$Q_{\text{vis}}(V)$  is physically meaningful within the bounds of  $\mathbf{V} = \mathbf{30\text{--}400\times}$ . At extremely low magnifications, the eye’s resolution limit dominates (CSF  $\approx 0$  across all telescope frequencies), leading trivially to  $Q_{\text{vis}} \rightarrow 1$ . Conversely, at exceptionally high magnifications ( $V > 500\times$ ), the CSF weighting shifts into the initial upward slope region of the CSF ( $< 3$  cpd), triggering a numerical artifact where  $Q_{\text{vis}}$  begins climbing again. To mitigate this, the program restricts the graph axis to  $400\times$ .

### 7.2 Model Boundaries

The model isolates and accounts exclusively for 4th-order spherical aberration (the  $\rho^4$  term). Coma, astigmatism, secondary mirror obstruction, and chromatic aberration (in catadioptric designs) are omitted. Additionally, the van Meeteren CSF models a statistical demographic average; individual ocular variation and specific dark adaptation states are not captured.

## 8 Conclusion

The visual quality index  $Q_{\text{vis}}(V)$  enables a nuanced, magnification-dependent evaluation of performance losses in spherical Newton mirrors. By blending a physically precise MTF model with the perceptual mechanics of the human eye’s CSF, it charts what an observer actually sees at a given magnification—offering a stark contrast to static, magnification-independent metrics like the Strehl ratio or pure MTF shape degradation.

The analytical approximation  $\tilde{Q}(V)$  serves as an effective tool for rapid estimation, mirroring correct qualitative trends ( $\tilde{Q} \rightarrow 1$  for  $V \ll V_{\text{krit}}$ ,  $\tilde{Q} \rightarrow S$  for  $V \gg V_{\text{krit}}$ ), though it introduces a systematic overestimation of perceived quality by 1–4% in mid-range magnifications. The

software plots both curves side-by-side to guarantee mathematical transparency regarding this model residual.

Finally, a major implementation defect—relying on the Maréchal approximation for  $W_b$  inversion—was resolved in v5.0.1 by implementing a rigorous bisection search inside the exact pupil integral. This historical bug artificially doubled the wavefront error  $W_b$ , causing highly exaggerated distortions along the user-controlled slider comparison curve.

## References

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